$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e \vec{u}_e) = 0 \qquad \frac{\partial \vec{u}_L}{\partial t} = \frac{e}{m_e} \nabla \Phi - \frac{1}{2} \nabla \left(\vec{u}_L + \frac{e\vec{A}}{m_e c} \right)^2 - \frac{\nabla p_e}{m_e n_e}$$

$$\vec{u}_e = \vec{u}_L' + \vec{v}_{os} \quad , \quad \vec{v}_{os} = \frac{e\vec{A}_0}{m_e c}$$

$$n_e = n_0 + n_e' \quad , \qquad \vec{A} = \vec{A}_L + \vec{A}' \qquad \Phi = \Phi'$$

$$\frac{\partial n_e'}{\partial t} + \nabla \cdot (n_0 + n_e') \left(\vec{u}_L' + \frac{e\vec{A}_0}{m_e c} \right) = 0 .$$

$$\Rightarrow \quad \frac{\partial n_e'}{\partial t} + n_0 \nabla \cdot \vec{u}_L' + \nabla n_e' \frac{e\vec{A}_0}{m_e c}$$